

Adaptations

ORGANISMS CHANGE OVER TIME IN ORDER TO SURVIVE.

Adaptations are the visible results of evolution. Natural selection alters the anatomy and behavior of organisms so they become adapted to new ways of living.

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Genetics I

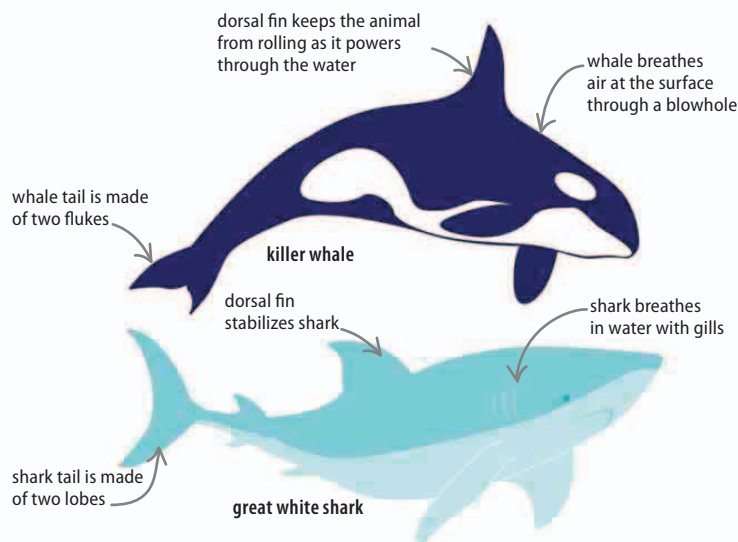
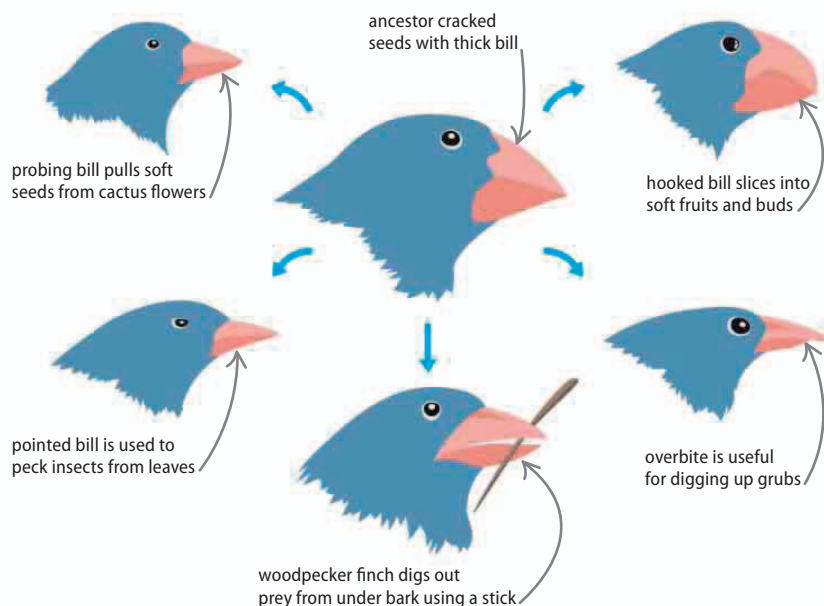
84–85▶

Adaptive radiation

When several different adaptations evolve from a single ancestor, it is known as adaptive radiation. The result is a group of species that share many features, but differ in ways that adapt them to a specific way of life. For example, rodents all have long, sharp incisors inherited from their common ancestor. However, in gophers these teeth are adapted to digging burrows, in beavers they fell trees, while squirrels use them to nibble through hard seed casings.

▷ Darwin's finches

Adaptive radiation can be seen in Darwin's finches—named after the discoverer of evolution. Most finches are seed-eaters, but the songbirds that live on the Galápagos Islands, Ecuador, have adapted to tackle other foods too.



Convergent evolution

A lot of evolution is divergent, with groups of related animals becoming less alike as they adapt to different environments. However, evolution can also be convergent, where unrelated species adapt to the same environment in the same way. For example, both birds and bats have evolved wings to enable them to fly. The shape, structure, and function of both kinds of wings are very similar, but birds and bats are only distantly related to each other, and their common ancestor did not have wings.

◁ Marine hunters

Sharks and toothed whales, such as dolphins and orcas, are all fast-swimming hunters. They look similar, but have very different body systems, because sharks are fish and whales are mammals.